LESSON 4.4

Find and Compare with Percentages

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the percent of a number using an equation or a model.

WHY THIS MATTERS It's Discount Day at the arcade! Different games are offering percent-off deals on ticket prices. The claw machine is 25% off its usual 60-ticket price, and the racing game is 40% off 80 tickets. You need to calculate the discounts to figure out how many tickets each game actually costs!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Percent *Spanish: Porcentaje*	A way to compare a number to 100, shown with the % sign.	45% = 0.45
Base *Spanish: Base*	The whole amount you start with — the number right after the word "of."	45% of 300 → base is 300
Part *Spanish: Parte*	How much of the base (the whole) you get. It is your answer.	45% of 300 is 135
Equation *Spanish: Ecuación*	A number sentence with an equal sign. The percent one is: part = percent × base.	part = 0.45 × 300
Discount *Spanish: Descuento*	Money taken off the first price to make it cheaper.	$40 shirt at 25% off: discount = $10
Double number line *Spanish: Recta numérica doble*	Two number lines that line up percents with amounts.	0%–100% over 0–300

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

140 tickets; 45% are student tickets. How many?

1. 45% = 0.45
2. 0.45 × 140 = 63

Answer: 63 student tickets

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Find 40% of 80. Step 1 — what is the base? (80, the number after “of”)

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base is _____ and the percent is _____ %.
- Written in hundredths, _____ % is _____ .
- My equation is part = _____ × _____ .
- The part is _____ , which is smaller than the base because _____ .

Word bank percent | base | whole | part | equation | decimal | hundredths | multiply | double number line | of | check | estimate

WATCH OUT!

A common mistake

A common mistake in Find the Percent of a Number is multiplying by the percent number itself instead of its decimal form

LESSON 4.4

Find and Compare with Percentages

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find each percent of a number using the formula: Part = Percent × Base.

Percent	Base	Part
25%	60	15
40%	80	
15%	200	

HINT The finished row shows the recipe: 25% of 60 became 0.25 × 60 = 15. Your rows use the same formula.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is 50% of 84?

(A) 34
(B) 42
(C) 50
(D) 168

HINT 50% is a friendly percent — it means exactly one-half of the number.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

3 **Circle the letter of the best answer. Show how you know.**

What is 10% of 320?

(A) 3.2
(B) 32
(C) 64
(D) 320

HINT 10% means one-tenth — there's a shortcut that needs no multiplication at all.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

4 **Circle the letter of the best answer. Show how you know.**

What is 20% of 150?

(A) 20
(B) 30
(C) 35
(D) 75

HINT '20% of 150' means you want a PART of 150 — the answer should be smaller than 150.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

5 **Circle the letter of the best answer. Show how you know.**

What is 40% of 50?

(A) 5
(B) 20
(C) 40
(D) 90

HINT 10% is one of 10 equal groups. First find 10% of 50.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Find 30% of 80** 30% of 80
2. **Multiply by the percent number** 30 × 80 = 2,400
3. **Final answer** 30% of 80 = 2,400

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** A percent is not a whole number — change 30% to a decimal before you multiply.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What steps do you follow every time you find the percent of a number?

SENTENCE STARTER To find ___% of ___, I first convert the percent to ___, then I multiply ___ × ___ = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 4.4

Find and Compare with Percentages

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

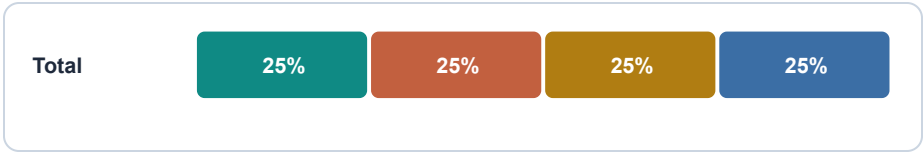
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PART 1 Understand the idea

models and meaning

1 **Use the model to solve. Show your work.**



200 ÷ 4 = 50. Each 25% section = 50 points. So 25% of 200 = 50.

SHOW YOUR WORK

Answer: _____

2 **Complete the table. Show how you found each value.**

Complete each row to find the percent of a number.

Percent	Base	Part
10%	250	
75%	120	
35%	200	
8%	50	

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

An arcade prize costs 240 tickets. If you have a coupon for 15% off, how many tickets do you save?

(A) 15 tickets

(B) 24 tickets

(C) 36 tickets

(D) 204 tickets



0% 50% 100%

WORK

4 **Circle the letter of the best answer. Show how you know.**

A game has 120 points. A player earns 35% of the points. How many points is that?

(A) 35

(B) 42

(C) 85

(D) 155



0% 50% 100%

WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Find 15% of 60** 15% of 60
2. **Change the percent to a decimal** 15% = 0.15
3. **Multiply by the wrong number** 0.15 × 15 = 2.25
4. **Final answer** 15% of 60 = 2.25

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What steps do you follow every time you find the percent of a number?

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LESSON 4.4

Find and Compare with Percentages

Challenge · Extension

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

A game costs $80. It is 25% off. What is the sale price?

(A) $20
(B) $60
(C) $55
(D) $100



0% 50% 100%

WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Percent Error

1. **Find 30% of 90** 30% of 90
2. **Convert percent to decimal** 30% = 3.0
3. **Multiply** 3.0 × 90 = 270
4. **Final answer** 30% of 90 = 270

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A class has 32 students. 25% join the math club and 50% join the reading club. How many students are in each club? How many more are in reading?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What steps do you follow every time you find the percent of a number?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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